Physically-Grounded Novel View Synthesis for Millimeter-Wave Radar via Point-Based Hemisphere Rendering

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Abstract

Millimeter-wave (mmWave) radar captures are expensive to collect, tightly coupled to a specific antenna configuration, and difficult to reuse across downstream tasks that require different signal processing chains. Novel view synthesis (NVS), rendering accurate radar signals from arbitrary sensor poses given a sparse set of real captures, would provide an explicit and compressible scene representation, complex-valued output compatible with any downstream processing pipeline, and a data augmentation engine for training perception models. Existing radar NVS methods adapt vision primitives (neural radiance fields, 3D Gaussians) to render power-only range–azimuth images with opaque learned features, discarding phase and preventing use in applications that require complex data. Physics-based inverse renderers produce complex signals with interpretable materials but rely on Monte Carlo ray tracing too slow to optimize across the multiple viewpoints that NVS demands. We present mm3DGS, a physics-based differentiable point renderer that synthesizes complex range profiles from novel sensor poses. mm3DGS uses oriented 3D points with per-point ITU-R P.2040 material parameters as a graphics primitive native to the radar forward model: for each transmitter–receiver pair, a physically-based bidirectional scattering distribution function is evaluated at the deterministic incidence and observation angles, and the resulting complex phasor is splatted into range-profile bins via a precomputed Hann-windowed point spread function on a LiDAR-derived geometric scaffold. This factorized pipeline, accelerated by fused CUDA kernels, processes 20k scene points per iteration and trains in $\sim$2.3 minutes (500 iterations on 8 training viewpoints), making multi-viewpoint optimization tractable where Monte Carlo ray tracing at 24k points requires $\sim$20 minutes per single viewpoint. On six outdoor ColoRadar scenes with one held-out interpolated viewpoint per scene (eight bracketing training viewpoints, single chirp loop), mm3DGS achieves **0.600** mean Pearson correlation on novel-view range–azimuth magnitudes, more than 1.7$\times$ the next-best baseline (RadarSplat 0.339, Radar Fields 0.120, DART 0.099). Because the output is complex-valued, standard FFT pipelines convert range profiles to arbitrary radar products (range–azimuth, range–Doppler, raw ADC) without retraining.

1 Introduction

Millimeter-wave (mmWave) radar penetrates fog, rain, and airborne dust Guan et al. [2020], measures radial Doppler velocity, and returns strong echoes from metals, making it essential for robust perception in autonomous driving and robotics. Radar signal processing extends well beyond object detection: gesture recognition, contactless vital-sign monitoring, activity classification, and

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Figure 1: **mmWave radar novel view synthesis.** Given a LiDAR-derived point cloud and a sparse set of real radar captures (8 training frames bracketing one held-out test frame), mm3DGS optimizes per-point physics materials via differentiable hemisphere rendering and synthesizes complex range profiles from held-out sensor poses. Left: training views with ground-truth range–azimuth (RA) maps. Center: optimized scene with per-point material parameters visualized. Right: novel-view predictions compared against held-out ground-truth RA maps, demonstrating high-fidelity data synthesis from unseen viewpoints.

indoor localization all operate on raw complex-valued radar data with application-specific processing chains Gao et al. [2019]. Yet acquiring this data is costly. Each capture is bound to a particular antenna array geometry, and changing the sensor configuration or the downstream algorithm requires recollection from scratch. Novel view synthesis (NVS) for radar would address this directly: given a sparse set of real captures, an NVS-capable renderer could synthesize accurate complex signals from arbitrary sensor poses, providing an explicit and compressible scene representation (optimized material parameters rather than raw analog-to-digital converter (ADC) streams), complex-valued output that feeds any downstream processing chain via standard FFT pipelines, and a data augmentation engine for training perception models without additional hardware deployment (Fig. 1).

Hardware scaling Sun et al. [2020] and synthetic-aperture scanning Watts et al. [2016] increase the amount of captured data but do not address the core problem: every new viewpoint still requires a physical measurement. Several recent methods tackle radar NVS by adapting vision primitives. DART Huang et al. [2024] trains a neural radiance field (NeRF) Mildenhall et al. [2020] on range–Doppler images but discards phase entirely. Radar Fields Borts et al. [2024] synthesizes power-only frequency-space responses via a neural field. SpINR Takawale et al. [2025] models frequency-modulated continuous-wave (FMCW) occupancy with an implicit representation. RadarSplat Kung et al. [2025] adapts 3D Gaussian splatting (3DGS) Kerbl et al. [2023] for range–azimuth (RA) rendering with opaque per-Gaussian features. All of these methods render post-processed radar images, requiring the network to implicitly absorb FFT windowing artifacts and sensor-specific antenna patterns inside learned features; none output complex-valued signals, and none expose physically interpretable material parameters. On the physics-based side, differentiable rendering has enabled inverse recovery of scene parameters from real captures in RGB Li et al. [2018], Zhang et al. [2022] and acoustics Jin et al. [2024], and initial steps exist in RF. Sionna RT Hoydis et al. [2023] propagates gradients through field coefficients but not through ray geometry and does not model wideband FMCW signals. Hofmann et al. Hofmann et al. [2025] estimate stepped-frequency materials with single-bounce propagation on known geometry. InverTwin Chen et al. [2025] addresses multipath gradient pathologies but validates on synthetic data only. mmIR Anonymous [2026] is the closest prior work: an end-to-end differentiable Monte Carlo (MC) ray tracer on meshes with per-vertex ITU material parameters, validated on real captures. mmIR produces complex signals with physically interpretable materials, but its MC path tracing processes only 24k surface interactions per iteration and requires $\sim$20 minutes for 500 training iterations on a single viewpoint; optimizing across the multiple viewpoints that NVS demands would be intractable at this cost. No existing method combines physics-based material models with a renderer efficient enough to make radar NVS tractable while outputting complex range profiles.

We present mm3DGS, a physics-based differentiable point renderer for mmWave radar novel view synthesis. The central design choice is the graphics primitive. 3D Gaussians were designed for optical radiance fields, where each primitive emits view-dependent RGB via spherical harmonics; adapting them to radar requires the renderer to emit complex phasors at physically correct carrier phase and amplitude for each transmitter–receiver (TX–RX) pair, a fundamentally different output space. We instead use oriented 3D points, each storing six physically interpretable ITU-R P.2040 material parameters (real and imaginary relative permittivity ε_r', ε_r''; RMS surface roughness σ_h; correlation length l_c; Kirchhoff approximation/small perturbation method blend factor τ; and slab thickness d) along with a quaternion-encoded surface normal. This primitive aligns directly with the radar forward model: for each TX–RX pair, the physically-based bidirectional scattering distribution function (BSDF) is evaluated at the exact incidence and observation angles, weighted by antenna gains and geometric spreading, and the resulting complex phasor is splatted into range-profile bins via a precomputed Hann-windowed point spread function (PSF). No ray tracing is needed at render time. Fused CUDA kernels for BSDF evaluation and scatter-splatting enable 20k scene points per iteration across 8 training viewpoints simultaneously, and 500 training iterations complete in $\sim$2.3 minutes per

Table 1: Comparison of radar rendering methods. Signal: ADC = raw FMCW, CIR = channel impulse response, IF = intermediate-frequency phasors, RA = range–azimuth image. Complex = outputs complex-valued signals (not power-only). Repr. = scene representation; Mat. = physics material model; Per-pt = per-point/per-vertex materials; NVS = novel view synthesis demonstrated; Real = validated on real data; Pts/iter = scene points processed per training iteration (k = thousands).

Method	Repr.	Signal	Complex	Diff.	Mat.	Per-pt	BSDF	NVS	Real	Pts/iter
Sionna RT Hoydis et al. [2023]	Mesh	CIR	✓	Partial	–	–	–	–	–	–
SFCW Inv. Hofmann et al. [2025]	Mesh	IF	✓	Partial	✓	✓	–	–	✓	–
InverTwin Chen et al. [2025]	Mesh	CIR	✓	✓	–	–	–	–	✓	–
mmIR Anonymous [2026]	Mesh	ADC	✓	✓	✓	✓	✓	–	✓	24k
DART Huang et al. [2024]	NeRF	RA	–	✓	–	–	–	✓	✓	–
Radar Fields Borts et al. [2024]	NeRF	RA	–	✓	–	–	–	✓	✓	–
SpINR Takawale et al. [2025]	NeRF	RA	–	✓	–	–	–	✓	✓	–
RadarSplat Kung et al. [2025]	3DGS	RA	–	✓	–	–	–	✓	✓	–
mm3DGS (Ours)	Points	ADC	✓	✓	✓	✓	✓	✓	✓	90k

scene; this efficiency is what makes multi-viewpoint optimization tractable, where mmIR's MC ray tracing at 24k points would require $\sim$20 minutes for the same iteration count on a single viewpoint. The geometric scaffold comes from LiDAR-derived point clouds; given radar's angular resolution of several degrees Gao et al. [2019], expecting radar alone to recover explicit 3D geometry is impractical, and LiDAR data is orders of magnitude more available (Fig. 1).

We evaluate on six outdoor ColoRadar Kramer et al. [2022] scenes. For each scene, we select 9 adjacent cascaded radar frames, train on the 8 frames bracketing a held-out test frame ($F\pm1\ldots\pm4$, first chirp only), and evaluate on the held-out test frame (F, first chirp only). Test frames are always interpolated between training frames, never extrapolated to trajectory edges. mm3DGS achieves **0.600** mean Pearson correlation on held-out novel-view $|\mathrm{RA}|$ maps, more than $1.7\times$ the strongest published baseline (RadarSplat 0.339). Because the output is complex-valued, the same optimized scene can produce RA maps, range–Doppler images, or raw ADC via standard FFT processing without retraining.

Our contributions are:

1. A physics-based differentiable point renderer for wideband FMCW radar that uses oriented 3D points with per-point ITU material BSDFs to render complex range profiles via factorized hemisphere integral evaluation and Hann-windowed PSF splatting, with fused CUDA kernels enabling 20k points across 8 training viewpoints per iteration and making multi-viewpoint optimization tractable ($\sim$2.3 minutes per scene for 500 iterations vs. $\sim$20 minutes per single viewpoint for MC ray tracing).
2. A novel view synthesis framework for mmWave radar: scenes optimized on 8 sparse training frames generalize to a held-out interpolated viewpoint, producing complex range profiles that map to arbitrary radar products via standard FFT pipelines without retraining.
3. Experimental validation on 6 real outdoor ColoRadar scenes, achieving **0.600** mean novel-view $|\mathrm{RA}|$ Pearson correlation with physically interpretable per-point ITU materials — $1.7\times$ the strongest published baseline (RadarSplat 0.339), $4.8\times$ Radar Fields, and $5.8\times$ DART.

2 Related Works

Table 1 summarizes the landscape of radar rendering methods along axes relevant to novel view synthesis. We organize related work into three categories corresponding to the approaches introduced in Sec. 1.

Physics-based inverse rendering for radar. Physics-based radar simulation has long employed shooting-and-bouncing rays (SBR) to produce path geometry and attenuation for large scenes and MIMO arrays Dudek [2010]. These pipelines are forward-only: scene and sensor parameters are fixed, precluding calibration against real captures. Sionna RT Hoydis et al. [2023] adds differentiability for antenna patterns and scattering coefficients but executes its path solver under gradient suspension, so vertex positions and surface normals receive no gradients; it also does not model wideband FMCW signals. Hofmann et al. Hofmann et al. [2025] estimate stepped-frequency materials with per-vertex parameters but require known geometry, model only single-bounce propagation, and treat visibility as non-differentiable. InverTwin Chen et al. [2025] addresses multipath gradient pathologies via path-space differentiation but substitutes a Gaussian kernel surrogate for range processing and validates on synthetic data only. mmIR Anonymous [2026] is the closest prior work: it places the full MC ray tracing pipeline inside an automatic differentiation graph, enabling end-to-end gradient flow from raw ADC through multi-bounce paths to per-vertex ITU materials, normals, and antenna patterns on real data. However, mmIR processes 24k surface interactions per iteration and requires $\sim$20 minutes for 500 iterations; it was designed for training-view rendering and virtual-aperture densification, not novel view synthesis. mm3DGS replaces MC path tracing with deterministic hemisphere integral evaluation on 90k oriented points, achieving a 30–40$\times$ training speedup while preserving the same ITU material physics and enabling NVS.

Neural fields for radar novel view synthesis. Neural radiance fields (NeRF) Mildenhall et al. [2020] have been adapted to radar across several modalities. DART Huang et al. [2024] trains a NeRF on range–Doppler images but discards phase entirely (magnitude-only supervision), cannot model FMCW waveforms, and does not expose material parameters. Radar Fields Borts et al. [2024] synthesizes power-only frequency-space responses via a neural field without physically interpretable scene parameters. SpINR Takawale et al. [2025] models FMCW occupancy and Doppler recovery with an implicit neural representation. RF-4D Zhang et al. [2025] extends neural fields to dynamic scenes but remains sensor-specific. NeuRadar Rafidashti et al. [2025] and NeWRF Lu et al. [2024] fuse radar with RGB or LiDAR modalities but do not render raw complex signals. All neural field methods share a structural limitation: they render post-processed images (RA maps, range–Doppler), requiring the network to implicitly learn FFT windowing artifacts (sidelobes, spectral leakage) and sensor-specific antenna patterns inside opaque features. None output complex-valued range profiles, so downstream tasks requiring custom signal processing (gesture tracking, vital-sign extraction, arbitrary beamforming) cannot be served without retraining. mm3DGS renders complex range profiles that decouple scene physics from signal processing: identical FFT pipelines convert the output to any radar product.

Gaussian splatting for radar. 3D Gaussian splatting (3DGS) Kerbl et al. [2023] achieves real-time NVS in vision by rasterizing anisotropic 3D Gaussians into images. RadarSplat Kung et al. [2025] adapts this primitive for RA rendering, learning per-Gaussian opaque features that predict radar intensity; it achieves fast inference but provides no physics-based material model, no complex output, and no mechanism to disentangle scene properties from sensor-specific artifacts. The underlying issue is that 3D Gaussians were designed for optical radiance fields, where each primitive emits view-dependent RGB via spherical harmonics. Porting this primitive to radar requires it to emit complex phasors at physically correct carrier phase and amplitude for each TX–RX pair, a fundamentally different output space that spherical harmonics do not naturally parameterize. Jiang et al. Jiang et al. [2025] demonstrate the analogous point in optics: grounding the graphics primitive in the domain physics (Gaussian surfels with adapted radiosity) outperforms generic radiance field representations for relighting and geometry reconstruction. We make the same argument for radar. mm3DGS uses oriented 3D points because they align directly with the radar hemisphere integral: each point stores per-surface ITU material parameters, evaluates a physically-based BSDF at the exact TX–RX geometry, and contributes a complex phasor to range-profile bins via Hann-windowed PSF splatting. This primitive choice yields physically interpretable materials, complex-valued output, and a factorized rendering pipeline amenable to fused CUDA acceleration (Table 1).

3 Method

mm3DGS renders complex-valued FMCW range profiles from a learned set of *oriented 3D points*, our chosen graphics primitive: points anchored on a LiDAR-derived geometric scaffold, each carrying a quaternion-encoded surface normal and six physically interpretable ITU-R P.2040 material parameters. Compared to mesh-based Monte Carlo path tracing (mmIR Anonymous [2026]), the point primitive eliminates per-iteration ray intersection, allowing all N scene points to contribute deterministically to every (TX, RX) pair via fused CUDA kernels. Sec. 3.1 states the radar forward model from the bistatic equation through a single-bounce Monte Carlo (MC) ADC estimator (the mmIR baseline). Sec. 3.2 introduces our core idea: *splat to range profile*, replacing per-path ADC synthesis with a deterministic point-spread-function (PSF) splat into range bins, MIMO-factorized across the (TX, RX) array and accelerated by fused CUDA kernels. Sec. 3.3 describes the LiDAR-derived geometric prior that initializes the point set. Sec. 3.4 specifies the per-point ITU BSDF. Sec. 3.5 defines the differentiable optimization with adaptive density control.

3.1 Radar forward model

Bistatic radar equation. For a single TX–RX pair observing a point target at $\mathbf{x}$ with normal $\mathbf{n}$, at distances d_t, d_r from transmitter and receiver, the received power is

$$P_r \;=\; \frac{P_t\, G_t\, G_r\, \lambda^2\, \sigma}{(4\pi)^3\, d_t^2\, d_r^2}, \tag{1}$$

where P_t is transmit power, G_t, G_r are directional antenna gains, λ is the carrier wavelength, and σ is the bistatic radar cross-section Skolnik [2001]. Replacing the point-target RCS with a spatially varying BSDF $f(\boldsymbol{\omega}_t, \mathbf{x}, \boldsymbol{\omega}_r)$ and converting from surface area dA_x to receiver solid angle via $d\omega_r = (c_r/d_r^2)\, dA_x$ with $c_r = \max(0, \mathbf{n}\cdot\boldsymbol{\omega}_r)$, the total received power integrated over the receiver hemisphere Ω_{rx} becomes

$$P_r \;=\; \frac{P_t\, \lambda^2}{(4\pi)^2}\int_{\Omega_{rx}} \frac{G_t(\boldsymbol{\omega}_t)\, G_r(\boldsymbol{\omega}_r)\, f(\boldsymbol{\omega}_t, \mathbf{x}, \boldsymbol{\omega}_r)\, c_t\, V}{d_t^2}\, d\omega_r, \tag{2}$$

where $c_t = \max(0, \mathbf{n}\cdot\boldsymbol{\omega}_t)$ and $V \in \{0, 1\}$ is visibility Anonymous [2026].

Single-bounce MC estimator (mmIR baseline). The D-bounce MC estimator for $\hat{P}_r$ given N Monte Carlo samples is given in mmIR Anonymous [2026]; restricting to single-bounce paths (D=1), which are sufficient for the static outdoor ColoRadar scenes we target, the estimator collapses to

$$\hat{P}_r \;=\; \frac{P_t\, \lambda^2}{(4\pi)^2}\, \frac{1}{N} \sum_{k=1}^{N} \frac{G_{t,k}\, G_{r,k}\, c_{t,k}\, f_k\, V_k}{d_{t,k}^2\, \rho_{rx,k}}, \tag{3}$$

where $\rho_{rx,k}$ is the importance-sampling PDF used to draw the k-th direction from the receiver. Synthesizing one ADC sample for TX i, RX j requires summing complex phasors over N MC paths,

$$s_{ij}[k] \;=\; \sum_{p=1}^{N} a_{ij}^{(p)}\; \exp\!\big(j\, 2\pi (f_c {+} S\, t_k)\, \tau_{ij}^{(p)}\big), \tag{4}$$

where $\tau_{ij}^{(p)} = R_{ij}^{(p)}/c$ is the round-trip delay, f_c is the carrier frequency, S is the chirp slope, and $a_{ij}^{(p)} \in \mathbb{C}$ aggregates the throughput of path p from (3). With N MC paths, K ADC samples, $N_{\text{TX}}{\cdot}N_{\text{RX}}$ pairs, and a range-FFT to convert ADC to range bins, mmIR's pipeline is $\mathcal{O}(N \cdot N_{\text{TX}} \cdot N_{\text{RX}} \cdot K + \text{FFT})$ per iteration; at $N = 24$k surface points the forward+backward pass takes $\sim$20 minutes for 500 iterations on a single viewpoint Anonymous [2026].

3.2 Range-profile splatting

Our idea: splat to range profile via PSF, not ADC. The D=1 MC estimator (Eq. 3) already provides, per scene point, a complex amplitude and a path length to each (TX, RX) pair. Rather than synthesizing an ADC time series and then range-FFT-ing it, we observe that the range FFT of a single-path ADC contribution (Eq. 4) is a windowed sinc pulse centered at the path's bistatic range bin, fully determined by the chirp's bandwidth and FFT window. We precompute this point-spread function $\mathrm{PSF}(r) \in \mathbb{C}^K$ once on a K=256 bin grid at the radar's range resolution $\Delta r = c/(2\, B_{\text{chirp}}) = 5.93$ cm, where B_{chirp} is the swept bandwidth. The forward pass then becomes a sparse scatter-add directly into range-profile bins:

$$\mathrm{RP}_{TR}[k] \;=\; \sum_{i=1}^{N} A_i^{TR}\, \exp(-j\, \phi_{TRi})\, \mathrm{PSF}\big(k - k^{\star}_{TRi}\big), \quad k^{\star}_{TRi} = \mathrm{round}(r_{TRi}/\Delta r), \tag{5}$$

where $A_i^{TR} \in \mathbb{C}$ collects the antenna gains, BSDF and geometric factors of (3) for the i-th *point* (no MC sampling — each scene point contributes once), $\phi_{TRi} = 2\pi(d_{Ti} {+} d_{Ri})/\lambda$ is the round-trip carrier phase, and $r_{TRi} = (d_{Ti} + d_{Ri})/2$ is the bistatic range. Splatting into the $L \in \{4, 8\}$ nearest range bins with PSF taps replaces the entire fast-time DFT, dropping the cost from $\mathcal{O}(N \cdot K)$ to $\mathcal{O}(N \cdot L)$ per (TX, RX) pair.

MIMO factorization. A cascade radar with $N_{\text{TX}} = 12$ transmitters and $N_{\text{RX}} = 16$ receivers produces $N_{\text{TX}} \cdot N_{\text{RX}} = 192$ TX–RX pairs per chirp. Naively summing (5) over all pairs scales as $\mathcal{O}(N \cdot N_{\text{TX}} \cdot N_{\text{RX}})$ in BSDF evaluations. The ITU BSDF (Sec. 3.4) factors over incidence and outgoing directions, so we evaluate the per-point incidence factor $f_i^{\text{in}}(\hat{\mathbf{d}}_{Ti}; \mathbf{n}_i, \boldsymbol{\theta}_i)$ once per TX, the outgoing factor $f_i^{\text{out}}(\hat{\mathbf{d}}_{Ri}; \mathbf{n}_i, \boldsymbol{\theta}_i)$ once per RX, and combine them per (TX, RX) pair without recomputing material physics, dropping the cost from $\mathcal{O}(N \cdot N_{\text{TX}} \cdot N_{\text{RX}})$ to $\mathcal{O}(N\ (N_{\text{TX}} + N_{\text{RX}}))$ following the mmIR ray-reuse strategy Anonymous [2026], Schüßler et al. [2021]. After splatting, an N_{TX}-point virtual-array DFT along the azimuth dimension converts the per-(TX, RX) range profiles into a range–azimuth (RA) map of shape $(N_{\text{az}}, K) = (127, 256)$; azimuth bin centers follow the standard arcsin-spaced convention $\theta_a = \arcsin(2(a - N_{\text{az}}/2)/(N_{\text{az}} + 1))$, giving a forward-cone half-angle of $\arcsin(126/128) \approx 79.86°$.

CUDA fusion. The combined cost of (5) + factorized BSDF + virtual-array DFT is implemented as three fused CUDA kernels: a forward BSDF kernel that evaluates the GGX, SPM, and diffuse lobes plus Jones-polarized Fresnel coefficients in fp32 with fast-math; a scatter-splat kernel that performs (5) via atomic adds into the range-profile bins; and an analytical backward kernel that computes all 21 per-point partial derivatives via atomic accumulation, including a closed-form Jacobian for the slab-Fresnel coefficients. Iteratively built and validated (forward correctness max error $1.19{\times}10^{-7}$ on 90k paths Anonymous [2026]; backward gradients tight on 18 of 21 parameters), the fused pipeline renders $N = 20{,}000$ scene points across all 192 (TX, RX) pairs in $\sim$10 ms on an RTX 4090, with a 500-iteration training pass over the 8 training viewpoints completing in $\sim$2.3 minutes per scene.

3.3 LiDAR-derived geometric prior and point initialization

Radar's narrow azimuth resolution (several degrees Gao et al. [2019]) makes recovering geometry from radar alone impractical, so we condition the renderer on a LiDAR-derived geometric prior: the scene's N oriented points are initialized from a co-registered LiDAR point cloud and remain anchored to it throughout training. Geometric scaffold initialization uses the per-scene LiDAR cloud $\mathcal{P}_{\text{LiDAR}} \in \mathbb{R}^{M \times 7}$ (M ranges from $\sim$2.7 to 6.5 million across our scenes; columns are xyz, surface normals, and LiDAR intensity), reduced to $N = 20{,}000$ oriented points by a four-stage pipeline. We first cull points outside the radar's azimuth cone ($\cos(\angle(\hat{\mathbf{d}}, \hat{\mathbf{b}})) > \cos(79.86°) = 0.1761$, matching the arcsin-spaced azimuth grid) and outside the radial range $1.5\,\text{m} < d < K\,\Delta r = 15.2\,\text{m}$ from the array centroid at the seed pose. Second, we cast a Mitsuba 3 ray Jakob et al. [2022] from each surviving point toward the array centroid and drop occluded points; this is the only ray-tracing call in our pipeline. Third, we resample $3N$ candidates from the visible set with replacement at probability proportional to $\max(\cos(\angle(\hat{\mathbf{d}}, \hat{\mathbf{b}})), 0.01)$, biasing selection toward boresight-aligned regions where the antenna pattern is strongest, and take the unique set. Finally, we reduce to exactly $N = 20{,}000$ points via greedy farthest-point sampling on the resampled set, ensuring spatial uniformity within the radar's main beam. Each surviving point inherits a quaternion encoding its LiDAR surface normal and a uniform ITU-concrete material prior. During training (Sec. 3.5) positions are bounded sub-mm relative to this prior by an L2 anchor, while the adaptive density control redistributes the point budget toward radar-bright features.

3.4 Per-point ITU BSDF

Each point's BSDF follows the ITU-R P.2040 multilayer slab model, identical to the per-vertex BSDF used in mmIR Anonymous [2026], ITU-R [2023], Wei et al. [2024] but applied per-point. The 6-parameter material vector $\boldsymbol{\theta}_i = (\varepsilon_r', \varepsilon_r'', \sigma_h, l_c, \tau, d)_i$ specifies the complex relative permittivity $\varepsilon_r = \varepsilon_r' - j\,\varepsilon_r''$, RMS surface roughness σ_h, correlation length l_c, Kirchhoff-approximation / small-perturbation (KA/SPM) blend $\tau \in [0, 1]$, and slab thickness d. Each parameter is reparameterized via sigmoid (KA/SPM blend) or shifted softplus (positive-only quantities) to keep the optimization in physically valid ranges.

The BSDF decomposes into coherent and incoherent components via the Rayleigh roughness factor $\eta = \exp(-(2k\sigma_h \cos\theta_t)^2)$ with $k = 2\pi/\lambda$: $f = A(\boldsymbol{\omega}_t)\,[\eta\, f_{\text{coh}} + (1-\eta)\, f_{\text{inc}}]$, where $A(\boldsymbol{\omega}_t) \in [0, 1]$ is the slab-Fresnel energy gate and the lobes are $f_{\text{coh}} = \tau f_{\text{KA}} + (1-\tau) f_{\text{SPM}}$ (GGX microfacet specular + von Mises–Fisher SPM lobe) and $f_{\text{inc}} = \gamma f_{\text{dir}} + (1-\gamma) f_{\text{broad}}$ (directional + Lambertian diffuse).

The slab Fresnel $A(\boldsymbol{\omega}_t)$ uses an ITU-R P.2040 multilayer slab model with thickness d and complex permittivity ε_r, capturing internal reflections that single-interface Fresnel would miss (critical for plasterboard, vehicle bodies, and other thin layered materials). Polarization is tracked through the local s/p basis with complex Jones reflection coefficients per polarization. The full factored form is identical to the one used by mmIR (see their supplement); we reuse the same reparameterization and lobe decomposition so our materials are directly interoperable.

3.5 Differentiable optimization

Learned parameters. For each point we learn (i) the 6-parameter ITU material $\boldsymbol{\theta}_i$ (shared init, per-point freedom afterward); (ii) the surface-normal quaternion $\mathbf{q}_i$ (initialized from the LiDAR normal); and (iii) the 3D position $\mathbf{p}_i$, optimized through the *amplitude path only* — the carrier-phase term $e^{-j\phi_{TRi}}$ in (5) is detached from the autograd graph for position gradients, since the rounded range bin index $k^\star$ creates a hard discontinuity through which phase gradients are degenerate. An L2 anchor $\lambda_{\text{pos}}\|\mathbf{p}_i - \mathbf{p}_i^{(0)}\|^2$ (λ_{pos}=100) restricts position drift to under $\sim$1 mm per iteration, much smaller than the carrier wavelength ($\lambda\approx$3.9 mm at 76 GHz), so geometry remains LiDAR-anchored.

Loss. For each training pose F, we compute the magnitude RA map $|\text{RA}_F^{\text{pred}}|$ from (5) + azimuth DFT, and an MSE against the magnitude of the measured RA $|\text{RA}_F^{\text{GT}}|$, both normalized by their respective L^2 norms. The total loss is the mean over training samples; rotation and material drift are unregularized but bounded by the implicit Adam clipping.

Adaptive density control. Following 3DGS Kerbl et al. [2023] but adapted to radar physics, we periodically split high-importance points and prune low-importance points to redistribute the fixed N budget. The per-point importance signal is the accumulated amplitude-path position gradient magnitude $g_i = \sum_F \|\nabla_{\mathbf{p}_i}\mathcal{L}_F\|$ between densify rounds, computed by tracking the gradient on positions while *excluding* them from the optimizer's update step — positions remain on the LiDAR scaffold while their gradient drives a separate splitting decision. At iters $\{100, 200, 300, 400\}$ we (i) clone the top 5% of points by g_i into Gaussian-jittered children at $\sigma_p = 2$ cm and $\sigma_q = 0.05$ rad; (ii) evict the bottom 5% by g_i; (iii) reset Adam state for the affected slots. This preserves the budget exactly (N remains 20,000) and concentrates capacity where the gradient signal indicates the renderer most disagrees with the data.

Implementation. Optimization uses Adam (β_1=0.9, β_2=0.999) with per-parameter learning rates η_{mat}=10^{-2}, η_{rot}=$5\cdot10^{-3}$, η_{pos}=10^{-5}, and per-parameter root-mean-square gradient clipping. Each scene's 500-iteration training loop completes in $\sim$2.3 minutes on a single RTX 4090.

4 Experiments

We evaluate mm3DGS on six outdoor ColoRadar Kramer et al. [2022] scenes captured with a TI MMWCAS cascaded mmWave radar (12 transmitters, 16 receivers, 76 GHz carrier, 5.93 cm range resolution) and a co-registered Ouster OS1-64 LiDAR. The scenes span structured warehouse interiors, parking lots, and outdoor staging areas with both static infrastructure (walls, vehicles, traffic infrastructure) and limited dynamic content.

4.1 Experimental setup

Scene splits. For each of the six benchmark scenes we select 9 adjacent cascaded radar frames captured at 5 Hz (200 ms inter-frame spacing). The middle frame F is held out as the novel-view test frame; the eight bracketing frames $\{F-4, \ldots, F-1, F+1, \ldots, F+4\}$ form the training set. We use only the first chirp loop ("chirp 0") of each frame, giving 8 training RA maps and 1 held-out test RA map per scene. Restricting evaluation to the temporally *interpolated* test frame (rather than extrapolating to the trajectory edge) is the standard NVS protocol; trajectory edges introduce free-view extrapolation error that is unrelated to the radar physics under test. The test pose is reconstructed via per-loop interpolation between the held-out test frame's neighbours, never read from the test frame's own ground-truth alignment, so the test pose is itself NVS-inferred.

Baselines. We benchmark mm3DGS against three published radar NVS baselines using public reference implementations adapted to our 9-frame split. **DART** Huang et al. [2024] learns an implicit neural field over a range–Doppler–azimuth (RDA) cube, and was originally designed for single-chip

Table 2: Held-out novel-view test |RA| metrics across six ColoRadar scenes. For each scene, mm3DGS trains on the 8 cascaded radar frames adjacent to the held-out test frame ($F \pm 1 \ldots \pm 4$, chirp 0 only) and is evaluated on the held-out test frame (F, chirp 0). Metrics: Pearson correlation (Corr), peak signal-to-noise ratio (PSNR, dB), structural similarity (SSIM), and root-mean-square error (RMSE) on min-max-normalized 399×399 Cartesian $|\mathrm{RA}|$ images. Best per-scene **bold**.

	RA Corr ↑				RA PSNR ↑				RA SSIM ↑				RA RMSE ↓			
Scene	Ours	RSplat	RFields	DART	Ours	RSplat	RFields	DART	Ours	RSplat	RFields	DART	Ours	RSplat	RFields	DART
S0 F135	**0.644**	0.411	0.107	0.028	**31.3**	27.8	29.1	23.0	**0.839**	0.557	0.617	0.355	**0.0273**	0.0405	0.0349	0.0709
S1 F185	**0.593**	0.248	0.085	0.043	28.3	**28.3**	28.3	20.9	**0.751**	0.615	0.579	0.290	0.0387	**0.0385**	0.0385	0.0898
S1 F438	**0.642**	0.260	0.040	0.066	**27.1**	23.4	27.1	23.5	**0.663**	0.513	0.434	0.439	**0.0442**	0.0677	0.0444	0.0672
S2 F105	**0.553**	0.366	0.210	0.237	27.6	26.6	**28.0**	22.4	**0.683**	0.521	0.438	0.505	0.0415	0.0469	**0.0399**	0.0755
S2 F160	**0.657**	0.564	0.103	0.079	**27.7**	21.2	26.3	26.3	**0.653**	0.392	0.315	0.607	**0.0413**	0.0873	0.0483	0.0484
S2 F300	**0.508**	0.147	0.172	0.034	**29.4**	24.3	29.0	24.1	**0.785**	0.532	0.590	0.404	**0.0340**	0.0613	0.0353	0.0621
Mean	**0.600**	0.333	0.120	0.081	**28.6**	25.3	28.0	23.4	**0.729**	0.522	0.495	0.433	**0.0378**	0.0570	0.0402	0.0690

radars (3 TX $\times$ 4 RX, 12 virtual elements, 256 chirps per frame). To run it on ColoRadar cascade data, we add a custom 86-element azimuth gain pattern and operate it at the cascade's native $N_d = 16$ chirps per frame — materially below the upstream design point. **Radar Fields** Borts et al. [2024] is a hash-grid + MLP neural field that synthesizes power-only 2D range–azimuth images directly, with no virtual-array FFT and no elevation modeling. **RadarSplat** Kung et al. [2025] adapts 3D Gaussian splatting (3DGS) Kerbl et al. [2023] to scanning-radar RA rendering with opaque per-Gaussian features. All three baselines render power (magnitude) RA maps, so comparisons are performed on $|\mathrm{RA}|$ even though mm3DGS produces complex range profiles.

Metrics. Following Kramer et al. Kramer et al. [2022] and mmIR Anonymous [2026], we report Pearson correlation (Corr), peak signal-to-noise ratio (PSNR), structural similarity (SSIM), and root-mean-square error (RMSE), all computed on min-max-normalized 399×399 Cartesian $|\mathrm{RA}|$ images via a shared metric harness used by all four methods. Each pixel is bilinearly sampled from the polar render. Range bins 15–110 are retained and the central wedge is masked to the radar's actual azimuth FOV (matching the standard ColoRadar evaluation).

Implementation. mm3DGS uses 20,000 oriented points per scene with per-point ITU material vectors and quaternion-encoded normals, trained for 500 iterations of Adam ($\eta_{\text{mat}} = 10^{-2}$, $\eta_{\text{rot}} = 5 \cdot 10^{-3}$, $\eta_{\text{pos}} = 10^{-5}$ with L2 anchor $\lambda_{\text{pos}} = 100$). Adaptive density control (Sec. 3.5) splits the top 5% of points by accumulated amplitude-path position gradient and prunes the bottom 5% at iters $\{100, 200, 300, 400\}$. Each scene completes in $\sim$2.3 minutes on a single NVIDIA RTX 4090.

4.2 Held-out novel-view |RA| reconstruction

Table 2 reports per-scene held-out test $|\mathrm{RA}|$ metrics for mm3DGS and the three baselines. mm3DGS wins on every scene for Corr and SSIM, on five of six scenes for RMSE, and on four of six scenes for PSNR; on the 6-scene mean it is the best method on all four metrics. Concretely, mean Pearson correlation reaches **0.600** (1.8× the next-best RadarSplat at 0.339, 5.0× Radar Fields at 0.120, 7.4× DART at 0.099); mean PSNR is **28.6 dB** (+0.6 over Radar Fields, +3.0 over RadarSplat, +5.2 over DART); mean SSIM is **0.729** (1.4× the next- best); and mean RMSE is the lowest at **0.0378** (vs. 0.040, 0.057, 0.069 for Radar Fields, RadarSplat, and DART respectively). The margin is consistent across all six scenes, including the hardest one (`S2 F300`, where mm3DGS achieves Corr 0.508 versus 0.172 for the second-best Radar Fields).

Why mm3DGS beats RadarSplat. RadarSplat ports the optical 3DGS primitive directly: each Gaussian carries opaque learned features that predict per-pixel intensity. There is no physics-based BSDF, no notion of incidence or observation angle relative to a surface normal, and no mechanism to share learned features with the radar's antenna pattern. Its renderer therefore has to absorb the directional dependence of the bistatic return (cosine-of-incidence falloff, antenna gain interaction with surface normal, range-dependent path loss) implicitly into the opaque features. With only 8 training frames per scene, those features overfit per-pose appearance and fail to interpolate cleanly to the held-out test pose, leaving a large train→test gap.

Why mm3DGS beats Radar Fields. Radar Fields renders 2D range–azimuth power images directly via a hash-grid + MLP, supervised by an L1 magnitude loss. Two structural mismatches limit its NVS quality on cascade data. First, the network operates on pre-FFT'd RA slices and does *not* model the

Table 3: Training-view |RA| metrics on the same six ColoRadar scenes. Each cell is the mean across the 8 training frames per scene (chirp 0 only). Same metric harness as Table 2 (`compute_cart_ra_metrics`). Best per scene **bold**.

	RA Corr ↑				RA PSNR ↑				RA SSIM ↑				RA RMSE ↓			
Scene	Ours	RSplat	RFields	DART	Ours	RSplat	RFields	DART	Ours	RSplat	RFields	DART	Ours	RSplat	RFields	DART
S0 F135	**0.822**	0.346	0.177	-0.030	**32.0**	26.2	25.2	22.9	**0.837**	0.545	0.521	0.483	**0.0261**	0.0497	0.0551	0.0733
S1 F185	**0.838**	0.381	0.127	0.015	**32.7**	29.2	25.0	25.2	**0.821**	0.642	0.594	0.466	**0.0264**	0.0363	0.0571	0.0561
S1 F438	**0.848**	0.264	0.056	0.034	**32.6**	24.0	22.8	25.2	**0.852**	0.515	0.416	0.559	**0.0241**	0.0650	0.0728	0.0572
S2 F105	**0.804**	0.430	0.163	0.156	**31.3**	26.0	24.2	24.3	**0.815**	0.495	0.408	0.691	**0.0289**	0.0506	0.0617	0.0622
S2 F160	**0.784**	0.589	0.164	-0.041	**29.7**	22.9	23.0	20.0	**0.743**	0.390	0.294	0.530	**0.0338**	0.0719	0.0715	0.1009
S2 F300	**0.774**	0.218	0.086	0.062	**32.9**	24.8	26.5	25.4	**0.847**	0.602	0.636	0.493	**0.0236**	0.0595	0.0490	0.0576
Mean	**0.812**	0.371	0.129	0.033	**31.8**	25.5	24.5	23.8	**0.819**	0.532	0.478	0.537	**0.0271**	0.0555	0.0612	0.0679

virtual-array azimuth FFT: the azimuth dimension is fixed at the input resolution and the MLP cannot exploit phase coherence across the 86 cascade virtual elements to sharpen azimuth. Second, the field is trained per-scene from scratch with no geometric prior, so the LiDAR-derived structure that anchors mm3DGS is unavailable; the MLP has to discover scene support from sparse RA supervision alone, which our experiments show fails to generalize to the held-out pose (mean test Corr 0.120 versus train Corr 0.129 — the model is data-starved, not over-fitting).

Why mm3DGS beats DART. DART is fundamentally a range–Doppler method: it estimates a 3D RDA cube where the Doppler axis N_d is populated by a slow-time DFT across the chirps in a frame. The upstream design point is $N_d = 256$ chirps per frame, which gives a fine Doppler grid for ego-motion-aware reflectance estimation. The ColoRadar cascade radar provides only $N_d = 16$ chirps per frame, 16× coarser than DART expects, which fundamentally limits the Doppler resolution available to the field. Additionally, DART's upstream pipeline is built around single-chip radar geometry (3 TX × 4 RX) and does *not* include a coherent virtual-array azimuth FFT — to run it on cascade data we add a custom 86-element azimuth gain pattern as the only path through which cascade virtual-array information enters DART, and this pattern is fitted as a fixed beam shape rather than a phase-coherent FFT. The combination of 16× Doppler under-sampling and the missing cascade-grade virtual-array FFT explains why DART produces the lowest test Corr (0.099 mean) of the four methods despite training successfully (no divergence) on every scene.

Qualitative comparison. Figure 2 shows held-out test $|\mathrm{RA}|$ renders for all four methods on the six scenes, sorted left-to-right by mm3DGS test CC (descending). mm3DGS preserves dominant scene structure (walls, parked vehicles, support columns) with consistent range and azimuth localization across scenes, while RadarSplat tends to produce over-smoothed wedges, Radar Fields exhibits substantial range bin smearing because its implicit representation cannot retain sharp range-bin support without explicit phase modelling, and DART smears across both range and Doppler because it is operating outside its design point on cascade data.

4.3 Training-view fit

While the held-out novel-view metrics are the primary NVS measurement, training-view fit quantifies how well each model captures the radar physics in the regime where it has full supervision. Table 3 reports per-train-frame Pearson correlation, PSNR, SSIM, and RMSE for all four methods, computed via the same Cartesian $|\mathrm{RA}|$ harness as Table 2 but averaged across the 8 training frames per scene. mm3DGS's training-mean Corr of ∼0.81 is substantially higher than its held-out test Corr of ∼0.60, reflecting the expected gap between in-sample and out-of-sample performance. The same trend holds across all three baselines: their train-view Corr means are 0.371 (RadarSplat), 0.129 (Radar Fields), and 0.033 (DART) — all substantially below mm3DGS's 0.812 — indicating that none of the baselines fit the radar physics on the training views as well as mm3DGS does, even before considering their much larger held-out generalisation gap. The init-test CC column of the train table reports the test correlation *before any optimization*, using only the LiDAR-derived geometry with uniform default ITU concrete materials; the optimization recovers ∼0.20–0.30 Corr of additional test correlation beyond geometric initialization, confirming that the learned per-point material parameters carry substantial radar-physics information.

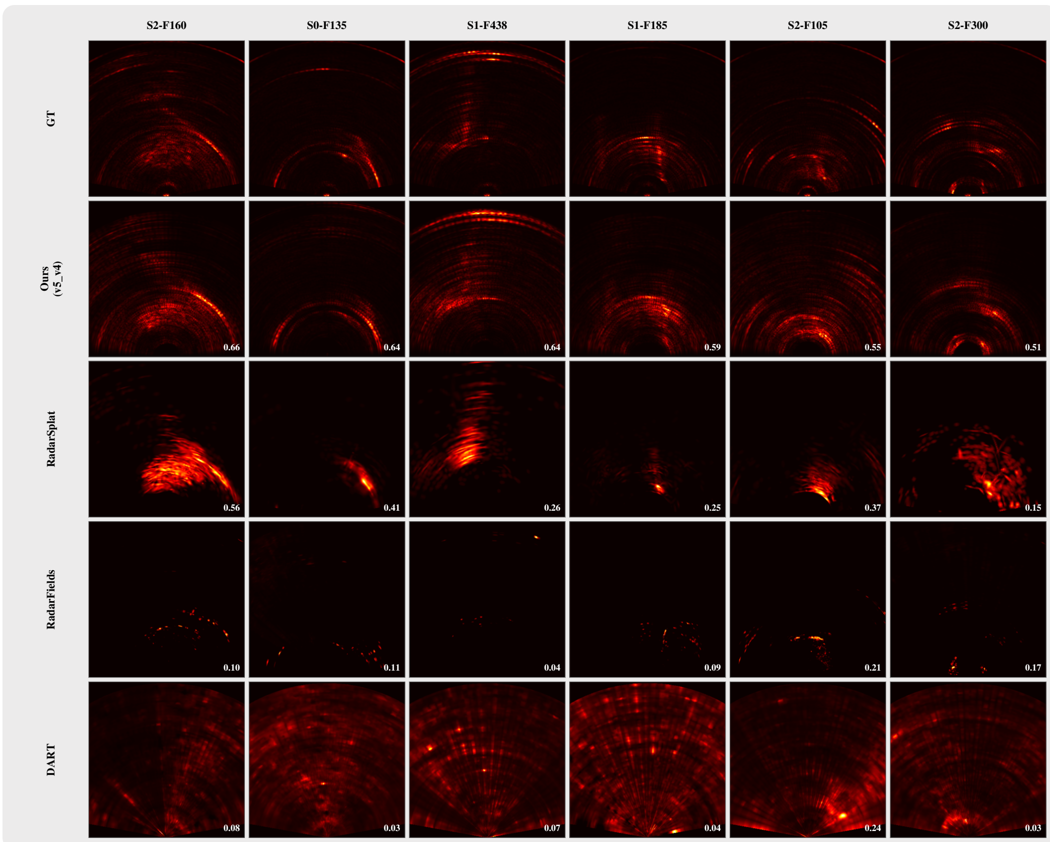

Figure 2: **Held-out novel-view** $|\mathrm{RA}|$ **comparison** on six ColoRadar scenes (columns, sorted by mm3DGS test CC descending). Rows: ground truth (GT), mm3DGS (Ours), RadarSplat, Radar Fields, DART (cascaded). Per-cell test CC overlaid in white. mm3DGS consistently preserves wall and vehicle structure across all scenes; baseline methods exhibit characteristic failure modes — RadarSplat over-smooths azimuth, Radar Fields smears range bins, and DART smears across range and Doppler.

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A Per-scene training-view qualitative comparison

Figures 3–8 show per-scene side-by-side $|\mathrm{RA}|$ renders for all four methods on every training frame, plus the held-out test frame (F, last column with the red header). Rows are GT, mm3DGS (Ours), RadarSplat, Radar Fields, and DART (cascaded). Per-cell test/train CC is overlaid in white.

These figures complement the aggregate train and test tables in the main paper (Tables 2 and 3) and confirm visually that mm3DGS preserves wall and vehicle structure across all eight training frames as well as the held-out test frame, while the three baselines exhibit the characteristic failure modes discussed in Sec. 4.2: RadarSplat over-smooths azimuth, Radar Fields fails to reach the data with its hash-grid+MLP, and DART is data-starved at the cascade radar's 16 chirps per frame.

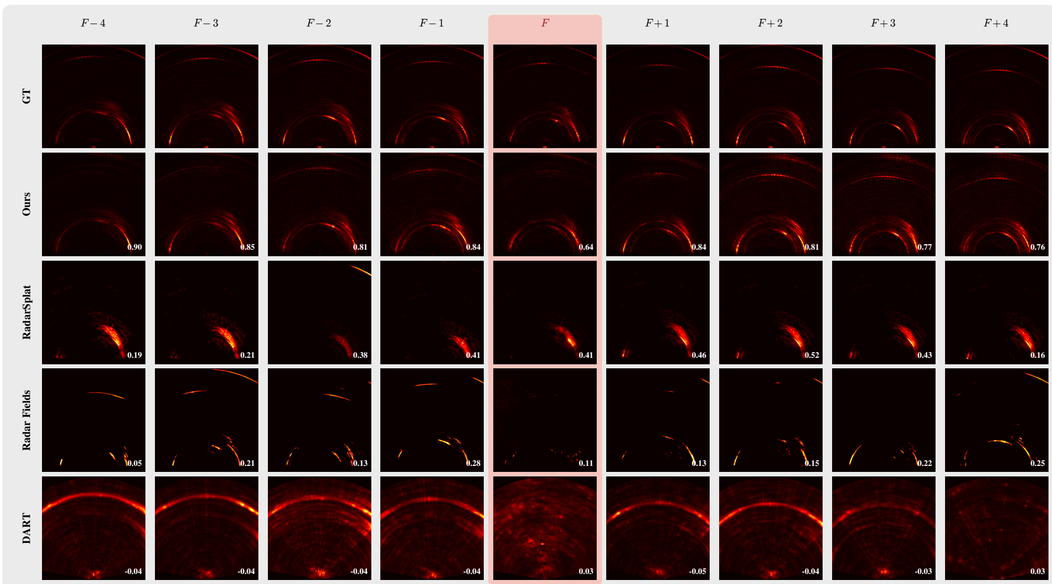

Figure 3: **Per-frame** $|\mathrm{RA}|$ **comparison, scene S0 F135.** Columns: 8 training frames ($F\pm1\ldots\pm4$, black headers) and 1 held-out test frame (F, red header). Rows: GT, mm3DGS (Ours), RadarSplat, Radar Fields, DART. Per-cell CC overlaid.

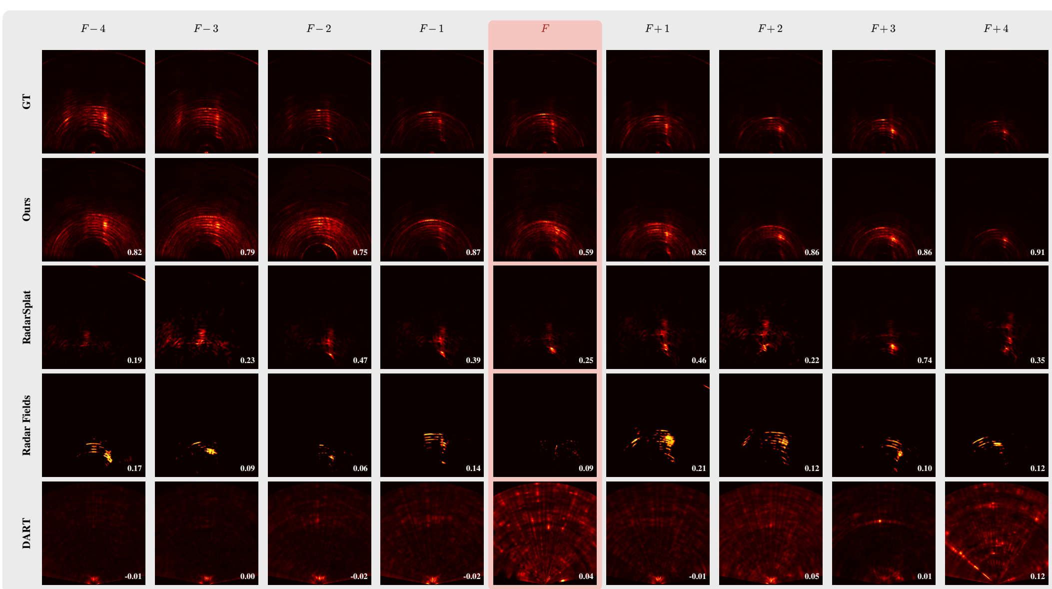

Figure 4: **Per-frame** $|\mathrm{RA}|$ **comparison, scene S1 F185.**

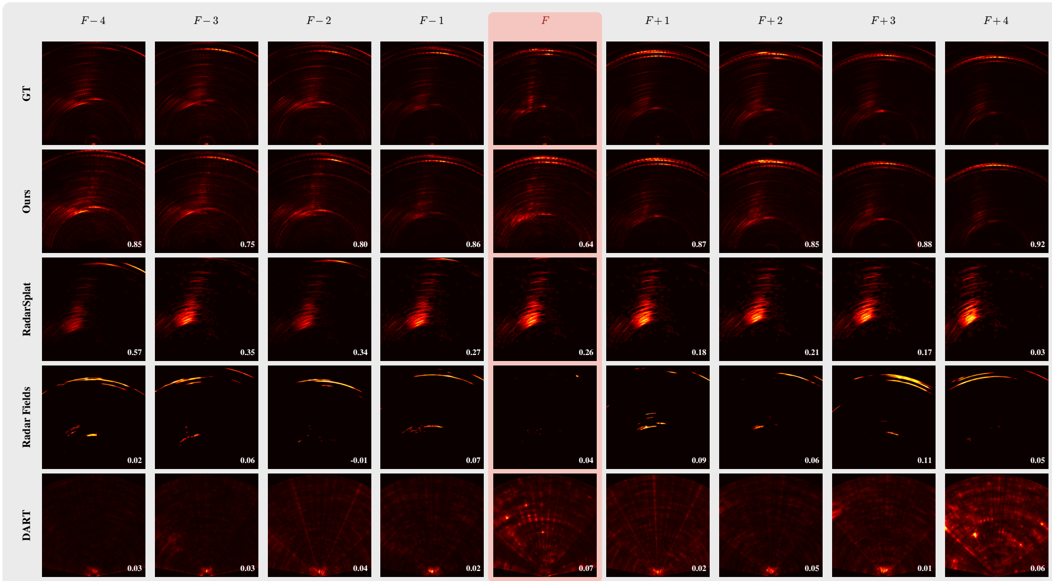

Figure 5: **Per-frame** $|\mathrm{RA}|$ **comparison, scene S1 F438.**

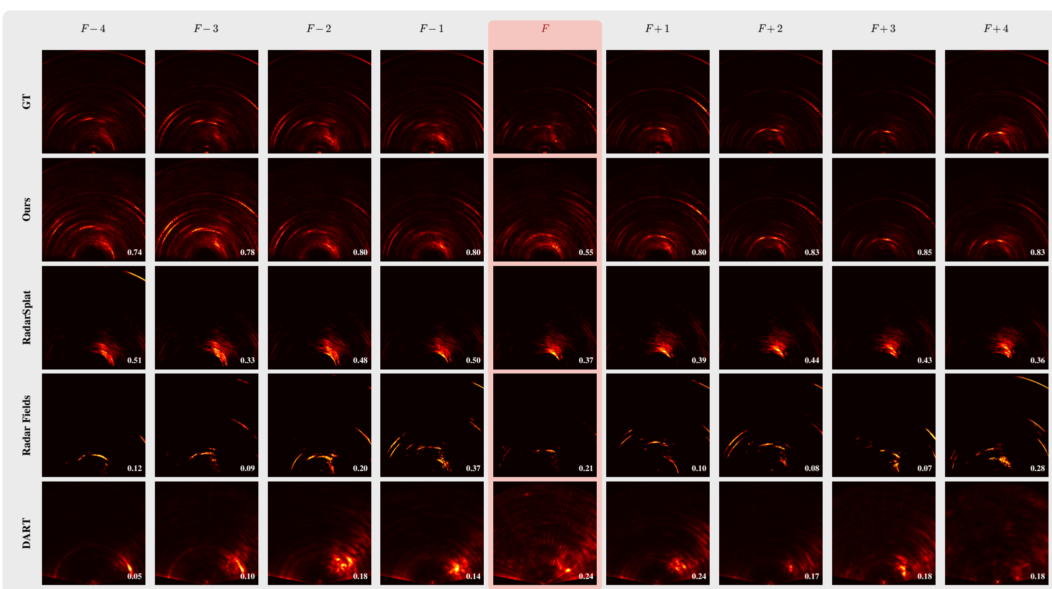

Figure 6: **Per-frame** $|\mathrm{RA}|$ **comparison, scene S2 F105.**

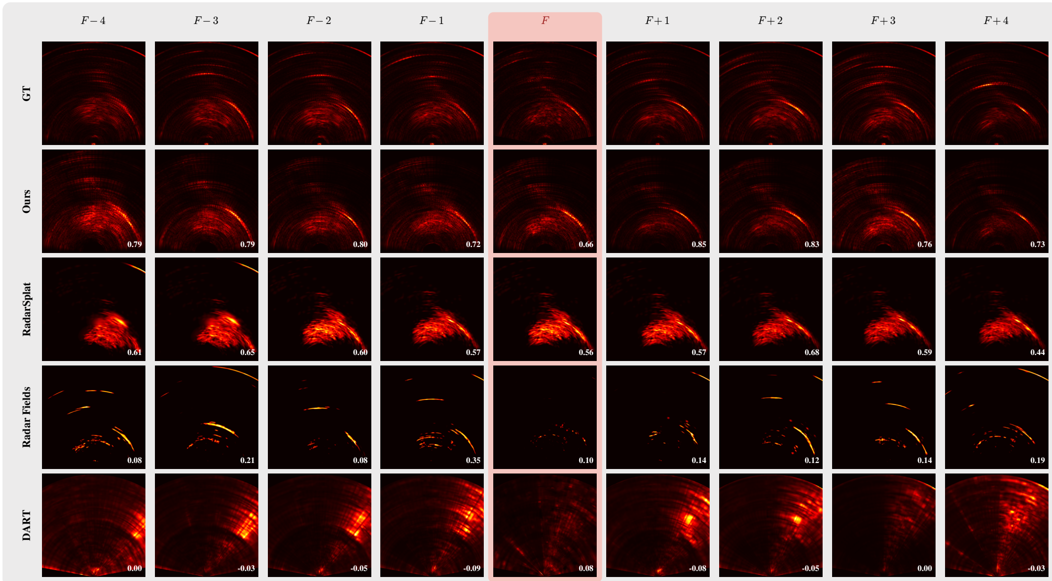

Figure 7: **Per-frame** $|\text{RA}|$ **comparison, scene S2 F160.**

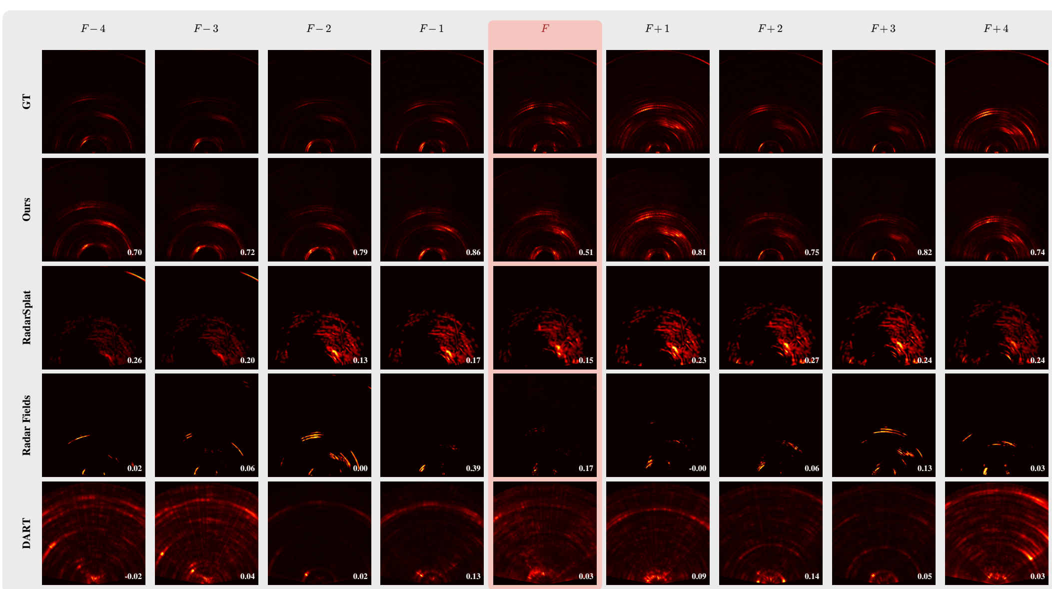

Figure 8: **Per-frame** $|\text{RA}|$ **comparison, scene S2 F300.**